

[DC] Visual Perception Enhancement via Transient Visual Cue in Immersive Virtual Reality

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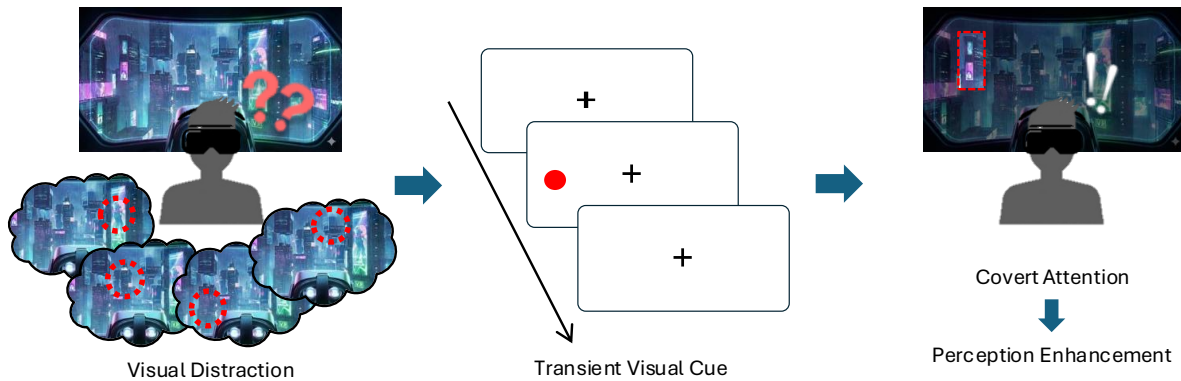


Figure 1: Perception enhancement without actual change of visual stimuli via transient visual cue

ABSTRACT

Virtual Reality (VR) offers users a realistic sense of presence by simulating environments rich in visual stimuli. To provide a practical and effective experience, it is essential to guide a user's behavior and perception within the VR environment as intended by the creator. Highlighting is a widely used method to increase perception of a specific object by making an object stand out from its surroundings, but it can disrupt the context or the immersive quality of the VR environment. This consortium paper presents research questions to address the issue and experimental plans to investigate a perception-enhancing method that preserves immersion.

Index Terms: Virtual Reality, Perception, Covert Attention

1 INTRODUCTION

Most existing Virtual Reality (VR) devices primarily provide visual information, offering users an immersive experience that feels like being in a virtual environment [3]. However, due to the human visual memory having a capacity limitation, the brain occasionally misperceives visual information [6]. To prevent the misperception of an essential element in a scene, methods are often used to highlight important aspects, making them stand out [5]; however, this may disrupt the context of the environment, and the immersiveness can be compromised.

A study conducted in a desktop environment found that a transient visual cue can enhance an observer's perception of an object at the cue's location [1]. I was inspired by this perception enhancement and plan to use this technique to investigate an approach for enhancing user perception of an arbitrary object in an immersive VR environment without breaking immersion.

Given that related studies were conducted in a desktop environment [1, 7], I plan to determine whether this perception enhancement by visual cues also occurs in VR and investigate whether the

visual cue can enhance perception even when it is located in a separate location from an object in a user's field of view. Lastly, I plan to investigate how an observer perceives the perception-enhanced object, and this method can help prevent breaking the immersion of the VR environment.

This is expected to aid in studying human perception in VR environments, and I hope it will provide a technical basis for enabling creators to present users with essential information effectively, as needed, without disrupting context or immersion.

2 MOTIVATION AND RESEARCH QUESTIONS

Vision is the most dominant sense among human senses, and because of this, most VR devices prioritize visual information, giving a user the feeling of being in a realistic virtual environment and enhancing immersion. However, cognitive limitations, such as the limited capacity of visual short-term memory (VSTM), limit the amount of information that can be processed simultaneously through vision [2].

The human brain tries to overcome this limitation by prioritizing critical visual information and efficiently using its limited capacity by highlighting a small number of elements that contrast with others and draw the user's attention. As a result, in VR environments with numerous objects, such as cities, it is challenging for users to identify the elements that the creator intends to prioritize. In this situation, enhancing a user's perception without actually changing the object can be a solution to this issue, rather than breaking the immersion of the VR environment by actually changing it [4]. A short transient visual cue (a simple black dot) presented in front of an object can cause the object to be perceived as having higher contrast [1] or moving faster [7] than the stimuli present actually. This enhancement raises the possibility that perception can be influenced without changing the visual information itself. However, most previous research on this topic has been conducted in 2-dimensional desktop displays, which have limited spatial freedom and depth.

Whether the same enhancement occurs in immersive VR, where users can experience an immersion and 3-dimensional depth, and if so, how much this perception changes, has not yet been studied. Therefore, I design studies by setting the following Research Questions to investigate how this modification occurs and how an

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observer perceives that modification in VR:

1. [Study 1] Does perception enhancement through visual cues also occur in immersive VR environments?
2. [Study 2] Does the placement of a visual cue in a field of view impact the perception enhancement?
3. [Study 3] Does an observer perceive this enhanced perception as an actual change in the object?

By addressing these questions, I aim to enhance our understanding of the process by which people perceive visual stimuli and establish a foundation for techniques that can guide a user's perception in a desired way without compromising context in the VR environment or reducing immersion.

3 RESEARCH PLANS

To address the research questions mentioned above, I have designed three experiments.

3.1 Study 1: Verifying Perception Enhancement in VR

This experiment replicates the phenomenon of the perception modification in a desktop environment, observed by Carrasco et al. [1], in an immersive VR environment. Here, I aim to investigate whether a visual cue-induced covert attention enhances contrast perception in VR. It also investigates whether this perception enhancement affects depth perception.

A participant is asked to select which of two objects with stripes has higher/lower contrast or is distant/near. Contrast and depth distance each have five levels, and one of the two objects is fixed at the middle level, while only the contrast or depth of the other object is adjusted based on the task conditions.

In each task, the participant sees a gray screen. Then, a visual cue (a black dot) appears on the left or right (horizontally, 15 degrees to the left or 15 degrees to the right of the center of the visual field) for a short time (67ms). After a short time (100ms), two Gabor patches appear on the left and right (at the same location where the visual cue appears) for a short time (44ms) and disappear. The participant has to select which one has a higher/lower contrast or is distant/near. The conditions for choosing are announced before the start of the task: Select the object with higher contrast / Select the more distant object. When comparing contrast, the depth remains the same; when comparing depth, the contrast remains the same.

The results are expected to demonstrate that covert attention induced by a visual cue can enhance contrast and depth perception in immersive VR, similar to desktop environments.

3.2 Study 2: Investigating spatial dependency

This study extends previous research to investigate how the position of the visual cue within a participant's field of view affects perception enhancement.

In this experiment, the visual cue is positioned at different locations, independent of the target objects being compared. The comparison target object is placed in the same location as in Study 1, and the visual cue appears at a random location in the near-peripheral or mid-peripheral region of the participant's field of view, depending on the condition. The experimental process is the same as the previous experiment.

This experiment will examine how the perception enhancement effect varies with the location of the visual cue, and it is also expected to examine how this effect varies with the spatial separation between the comparison target object and the visual cue.

3.3 Study 3: Is enhanced perception recognized as a change?

In this experiment, a participant will be asked to identify objects that have different contrasts or are located at different depths between two sequentially presented similar scenes.

The participant is shown a few objects in the first scene, then a gray surface for a short time. Then, a scene similar to the last scene is shown. One object in the second scene has a different contrast or is located at a different depth compared to the last scene, or nothing has changed. The participant is then asked to select the object that has changed. Just before the second scene is presented, depending on the condition, a visual cue is given for a short time to enhance the perception of one object (the changed object or the unchanged object). The participant can select the object that they think has changed or respond that no object has changed. I will examine how participants' choices change across each change condition by varying contrast and depth differences.

I expect that this experiment can investigate whether people perceive the enhanced perception as a change in the object. In addition, I expect this investigation to determine how significant a change in contrast or depth is needed to achieve the same level of perception enhancement.

Since this perception modification in VR has not yet been investigated, I plan to determine specific conditions, such as the number of objects provided in one scene or the duration of the gray screen, based on the results of previous experiments and pilot studies.

4 CONCLUSION

Through the three experiments presented above, I aim to verify whether the perception enhancement induced by a visual cue occurs in an immersive VR environment. And, I will analyze how this modification changes depending on the location of the visual cue and how people perceive it.

If this perception enhancement occurs in a VR environment and the experimental results are consistent with the expectation that people perceive enhanced and non-enhanced perception differently, I will propose a new method that induces the perception of an unchanged object that seems to have changed. This serves as a complementary concept to change blindness, where users fail to recognize actual changes.

Additionally, I believe the three experiments above will help study people's perception of change within a VR environment, and they will provide a technical basis for controlling a user's perception of a specific point as the creator wants, without breaking the immersion or the context of the environment.

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